TEST CODE **000592**

FORM TP 9795

MAY/JUNE 1997

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

MATHEMATICS

Paper 02 - General Proficiency

2 hours 40 minutes

04 JUNE 1997 (a.m.)

INSTRUCTIONS TO CANDIDATES

1. **Answer ALL questions in Section I, and any TWO in Section II.**
2. **Begin the answer for each question on a new page.**
3. **Full marks may not be awarded unless full working or explanation is shown with the answer.**
4. **Mathematical tables, formulae and graph paper are provided.**
5. **Mathematical instruments, slide rules and silent electronic calculators may be used for this paper.**
6. **You are advised to use the first 10 minutes of the examination time to read through this paper. Writing may begin during this 10-minute period.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

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SECTION I

Answer ALL the questions in this section.

1. (a) Calculate the exact value of

(i) $\dfrac{2.8 + 1.36}{4 - 2.7}$

(ii) $\left(\dfrac{27}{8}\right)^{\frac{1}{3}}$ **(6 marks)**

(b) Calculate 9.72 x 12.05 and write your answer

(i) exactly

(ii) correct to 2 decimal places

(iii) correct to 2 significant figures

(iv) in standard form. **(4 marks)**

Total 10 marks

2. (a) If $\dfrac{3x + 1}{3} - \dfrac{x - 2}{2} = 2 + \dfrac{2x - 3}{3}$,

find the value of x. **(5 marks)**

(b) Factorize completely:

(i) $15x^2y - 20xy^2$

(ii) $3 - 12b^2$ **(4 marks)**

(c) Given that $m = -3$, $n = 2$, $p = -1$,

find the value of $\dfrac{m(p - n)^2}{3p + m}$. **(4 marks)**

Total 13 marks

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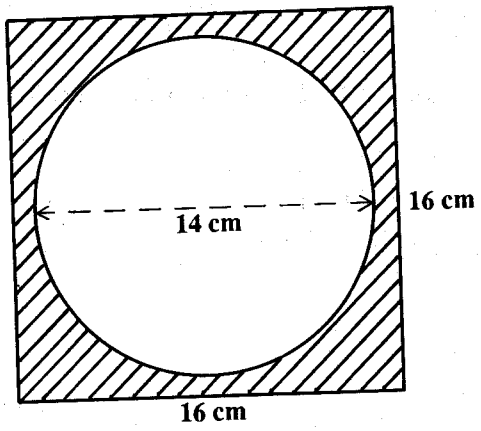
3.



A rectangular block of wood is 6 cm high. The cross-section of the block is a square of side 16 cm. A cylindrical container is carved out of the block. The cylinder is 5 cm deep and the diameter of the cross-section is 14 cm. The figure above, **not drawn to scale**, shows the top surface of the container.

Calculate

(a) the volume of the rectangular block of wood **(2 marks)**

(b) the area of the wooden top surface of the container **(3 marks)**

(c) the capacity, in litres, of the container **(3 marks)**

(d) the volume of wood in the container. **(2 marks)**

(Use $\pi = \frac{22}{7}$) **Total 10 marks**

4. (a) The charges for electricity are given in the table below:

ITEM	RATE
Rental	$2.00 per month
Energy charge	15¢ per KWH
Fuel charge	2.05¢ per KWH

The meter readings for the KWH used for December and January were:

Previous Reading 30 November	Present Reading 31 January
05783	06593

A government tax of 15% of the total charges is added to the bill.

Calculate

(i) the number of KWH units used

(ii) the total amount to be paid. **(7 marks)**

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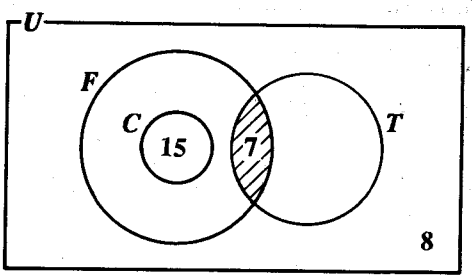
(b)



The Venn diagram above represents information about the 40 members of a youth club.

F = {members who play football}
C = {members who play cricket}
T = {members who play tennis}

The same number, x, play football only and tennis only.

(i) Calculate the number who play football.

(ii) State the information represented by the shaded region of the Venn diagram.

(iii) State the relationship between the members of C and F, and between C and T.
(5 marks)

Total 12 marks

5. f and g are functions defined as follows:

$f : x \rightarrow 3x - 5$

$g : x \rightarrow \frac{1}{2}x$

(a) Calculate the value of $f(-3)$. **(2 marks)**

(b) Write expressions for (i) $f^{-1}(x)$

(ii) $g^{-1}(x)$. **(4 marks)**

(c) Hence, or otherwise, write an expression for $(gf)^{-1}$. **(4 marks)**

Total 10 marks

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6. (a) (i) **Using ruler and compasses only,** construct the trapezium $KLMN$ with KL parallel to NM, $KL = 4$ cm, $NM = 7$ cm, $\angle KLM = 120°$ and $LM = 6$ cm.

(ii) Show that $\angle NKL = \angle MNK = 90°$.

(iii) Show, by calculation, that the distance between KL and MN is $3\sqrt{3}$ cm.

(9 marks)

(b) A plane takes off at an angle of elevation of 17° to the ground. After 25 seconds the plane has travelled a horizontal distance of 2400 metres.

(i) Draw a sketch to represent the given information.

(ii) Calculate to 2 significant figures the height of the plane above the ground after 25 seconds. **(4 marks)**

Total 13 marks

7. (a) In a competition, 3 points are awarded for winning a match, 0 points if the match was lost, and 1 point for a draw.

The results of seven matches played by three teams, J, K and L are:

Team J won 3, lost 1, and drew 3; Team K won 2, lost 4, and drew 1; Team L won 3, lost 2, and drew 2.

(i) Write a column matrix to represent the points awarded for win, lose and draw. Name the matrix P.

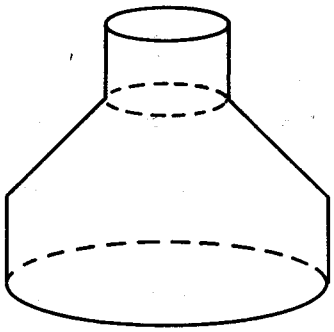
(ii) Write a matrix to represent the results of the matches played by the three teams. Name the matrix R.

(iii) Construct a matrix product which may be used to find the number of points each team was awarded at the end of the competition.

(iv) Using your matrix product, calculate the total points awarded to each team.

(5 marks)

(b)



The figure above, **not drawn to scale**, is a sketch of a symmetrical jar. The jar is in the shape of two cylinders joined by a frustum of a cone. The diameter of the bottom cylinder is 8 cm and of the top cylinder is 2 cm. The height of each cylinder is 2 cm. The total height of the jar is 8 cm.

Construct full-size drawings of

(i) the plan

(ii) an elevation. **(7 marks)**

Total 12 marks

8. An answer sheet is provided for parts (b) and (c) of this question.

The table below shows the distribution of 100 pieces of ribbon by length.

Length (cm)	No. of pieces	Cumulative Frequency
1 – 10	3	3
11 – 20	7	10
21 – 30	10	
31 – 40	19	
41 – 50	33	
51 – 60	14	
61 – 70	8	
71 – 80	6	

(a) Copy and complete the cumulative frequency column in the table in your answer book. **(2 marks)**

(b) **On the answer sheet provided**, use a scale of 2 cm to represent 10 cm on the x-axis, and 2 cm to represent 10 pieces of ribbon on the y-axis. Draw a cumulative frequency curve to represent the information given in the table. **(4 marks)**

(c) Using your graph, estimate

(i) the lower quartile

(ii) the upper quartile. **(2 marks)**

(d) A piece of ribbon is chosen at random. Calculate the probability that it is NOT longer than 50 cm. **(2 marks)**

Total 10 marks

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SECTION II

Answer TWO questions in this section.

RELATIONS AND FUNCTIONS

9. An answer sheet is provided for part (b) of this question.

(a) (i) Sketch the graph of the function

$$F = x^2 + x - 6 \quad \text{for} \quad -4 \leq x \leq 3.$$

From the graph, estimate for the domain given

(ii) the range of F

(iii) the interval of the domain for which

$F(x) < 0.$ **(7 marks)**

(b) The figure on the graph sheet shows the graph of the curve $y = 2x^2 - 3x - 2$.

(i) Using the graph, find the gradient of the curve at the point where $x = 2$.

(ii) On the same axes, draw the graph of $5y = 18x + 36$.

(iii) Using your graphs, find the solution of the simultaneous equations

$$y + 2 = 2x^2 - 3x$$
$$\text{and} \quad 5y = 18x + 36.$$

(8 marks)

Total 15 marks

10. An answer sheet is provided for this question.

The diagram on the graph sheet represents the distance-time graphs of a bus and a car.

The bus leaves a town X at 08:30 hrs and travels to another town Y. Its distance-time graph is $DFGH$. The car leaves Y and travels to X. Its distance-time graph is $PGQR$. There is a check-point M between X and Y where all vehicles MUST stop.

Use the graphs to answer the following.

(a) State

(i) the distance between X and Y

(ii) the position of M. **(3 marks)**

(b) What information about the vehicles is given by

(i) FG?

(ii) GQ?

(iii) G? **(3 marks)**

(c) At what time did the car

(i) leave Y?

(ii) reach M?

(iii) arrive at X? **(3 marks)**

(d) For the journey between Y and M, calculate the velocity in kmh^{-1} of

(i) the bus

(ii) the car. **(4 marks)**

(e) Calculate the average speed of the car for the whole journey to the nearest kmh^{-1}. **(2 marks)**

Total 15 marks

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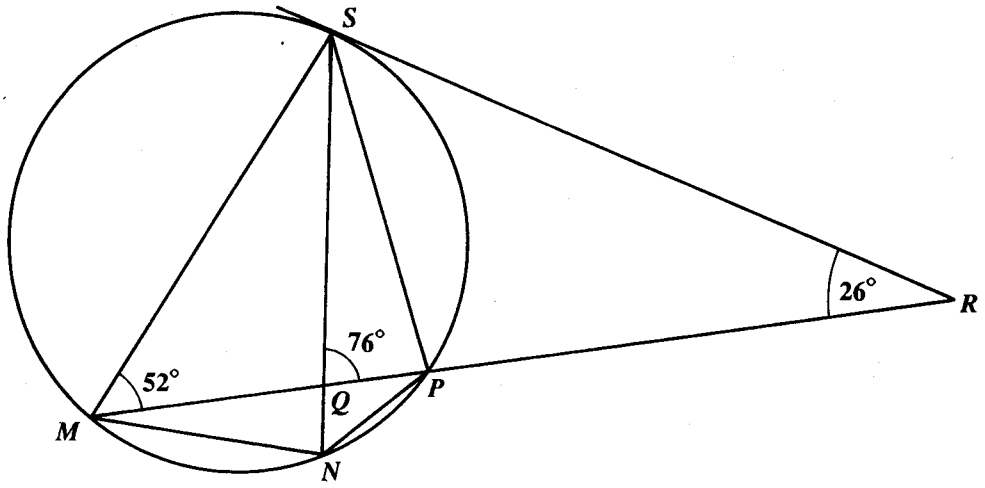
TRIGONOMETRY AND GEOMETRY

11.



In the figure above, **not drawn to scale**, SR is a tangent to the circle. The chords SN and MP intersect at Q. The chord MQP produced meets the tangent SR at R. Angle $SRP = 26°$, angle $QMS = 52°$ and angle $PQS = 76°$.

(a) Calculate, giving reasons

(i) angle MSQ

(ii) angle RSP

(iii) angle SPN. **(10 marks)**

(b) Given that $SR = 12$ cm, show that the area of $\Delta MRS = 39$ cm^2, correct to 2 significant figures. **(5 marks)**

Total 15 marks

GO ON TO THE NEXT PAGE

12. **In this question, use $\pi = 3.14$**

(a) A wheel is turning at the rate of 33 revolutions per minute. Express this speed in radians per second, correct to 2 significant figures. **(3 marks)**

(b) An airport is situated at G (40° N, 50° W). Another airport at F lies east of G. The airports at G and F are 3×10^3 km apart, measured along the circle of latitude.

(i) Calculate the length of the circle of latitude 40° N, expressing the answer in standard form.

(ii) Calculate the longitude of F.

(iii) Another airport is situated at H(50° N, 130° E). Calculate the shortest distance between H and G, expressing the answer in standard form.

[Use radius of the earth = 6370 km] **(12 marks)**

Total 15 marks

VECTORS AND MATRICES

13. $KLMN$ is a quadrilateral. The position vector of K is $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$, of L is $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and of M is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

(a) Find (i) $\overrightarrow{LM}$

(ii) $\overrightarrow{KM}$. **(4 marks)**

(b) Write the equation of the line

(i) LM

(ii) KM. **(2 marks)**

(c) P is the midpoint of KM. Find the position vector of P. **(3 marks)**

(d) P is also the midpoint of LN. Show that the position vector of N is $\begin{pmatrix} -2 \\ -2 \end{pmatrix}$.

(3 marks)

(e) Show that $KLMN$ is a parallelogram, giving reasons. **(3 marks)**

Total 15 marks

14. An answer sheet is provided for part (a) of this question.

(a) The position vector of H is $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$, of K is $\begin{pmatrix} 1 \\ 5 \end{pmatrix}$, and of W is $\begin{pmatrix} -3 \\ 3 \end{pmatrix}$. H, K and W are the vertices of ΔHKW. The triangle HKW is reflected in the line $y = -x$.

(i) State the position vectors of the vertices of the image.

(ii) Using 2 cm to represent 1 unit on each axis, draw on graph paper the triangle HKW and its image.

(iii) Name the axes of symmetry of the shape made by the object and the image together.

(9 marks)

(b) The matrix $N = \begin{pmatrix} 1 & 1 \\ -1 & -2 \end{pmatrix}$ describes a transformation on ΔGDF. The coordinates of the image are $G'(3, 2)$, $D'(5, 2)$ and $F'(4, 3)$.

Find

(i) the matrix which would transform $\Delta G'D'F'$ onto ΔGDF

(ii) the coordinates of G, D and F. **(6 marks)**

Total 15 marks

END OF TEST

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MATHEMATICS

Paper 02 - General Proficiency

Answer sheet for Question 8 (b) & (c). **Candidate number .....**

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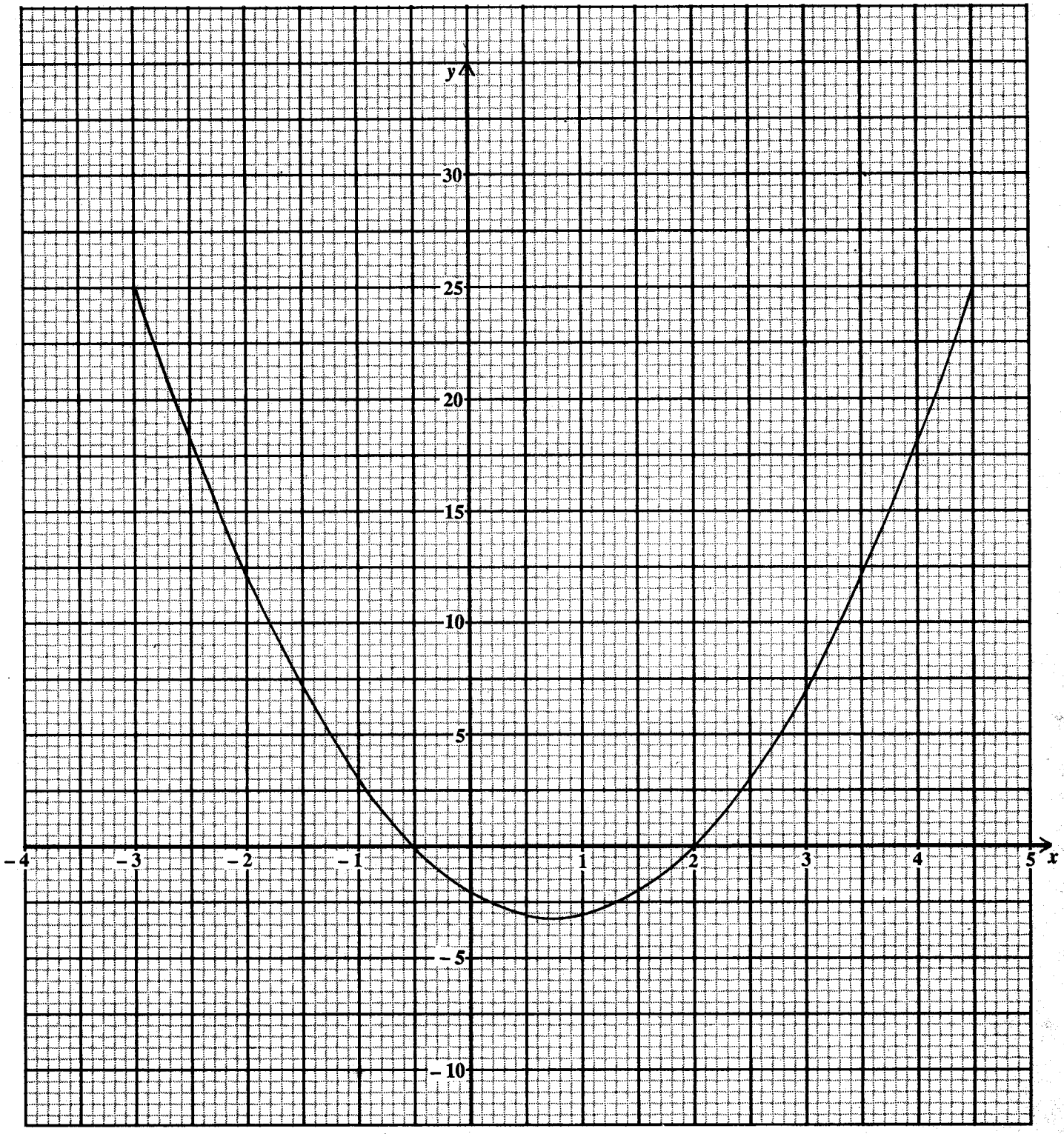
SECONDARY EDUCATION CERTIFICATE EXAMINATION

MATHEMATICS

Paper 02 - General Proficiency

Answer sheet for Question 9 (b). **Candidate number** .....



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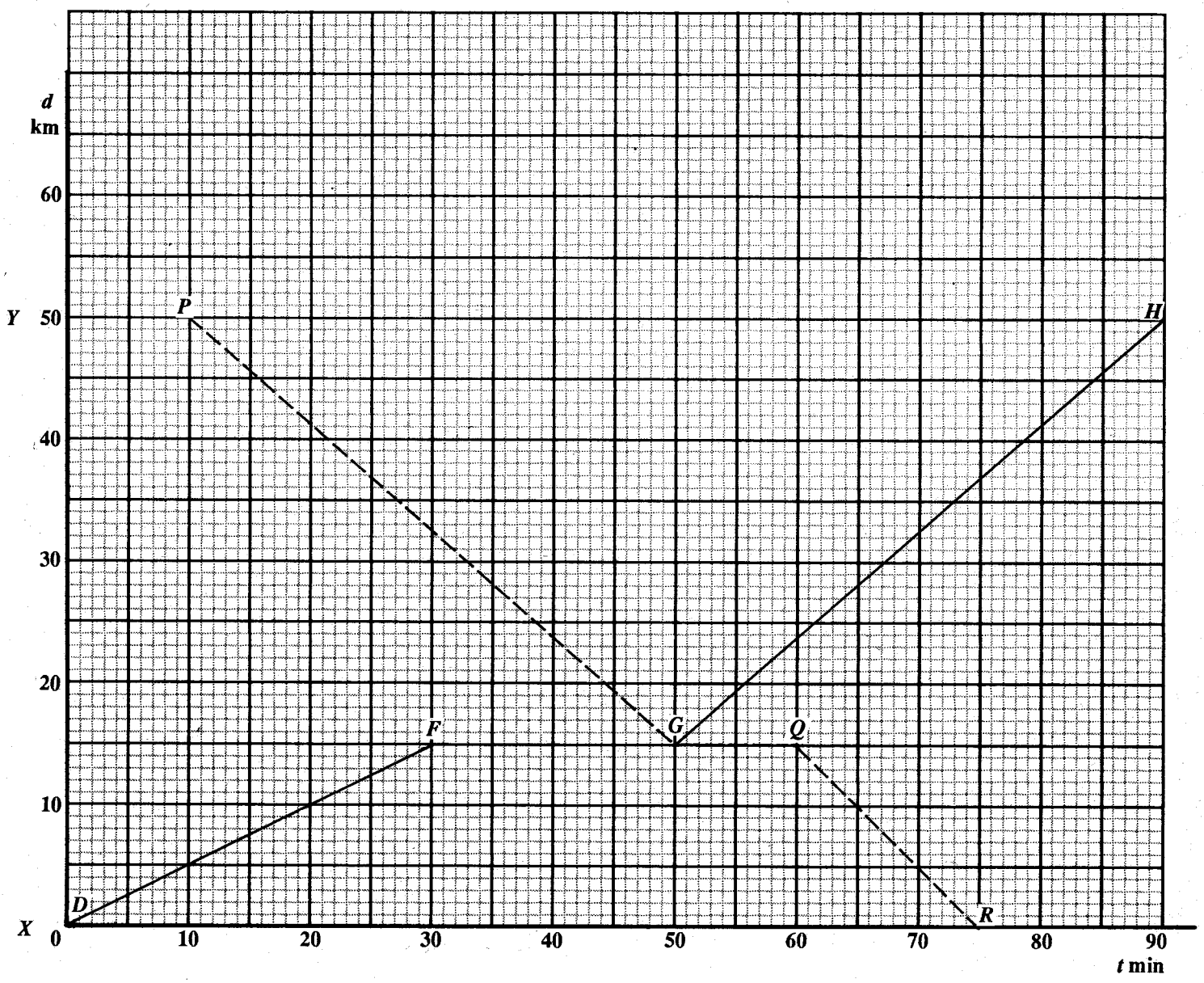
MATHEMATICS

Paper 02 - General Proficiency

Answer sheet for Question 10. **Candidate number .....**



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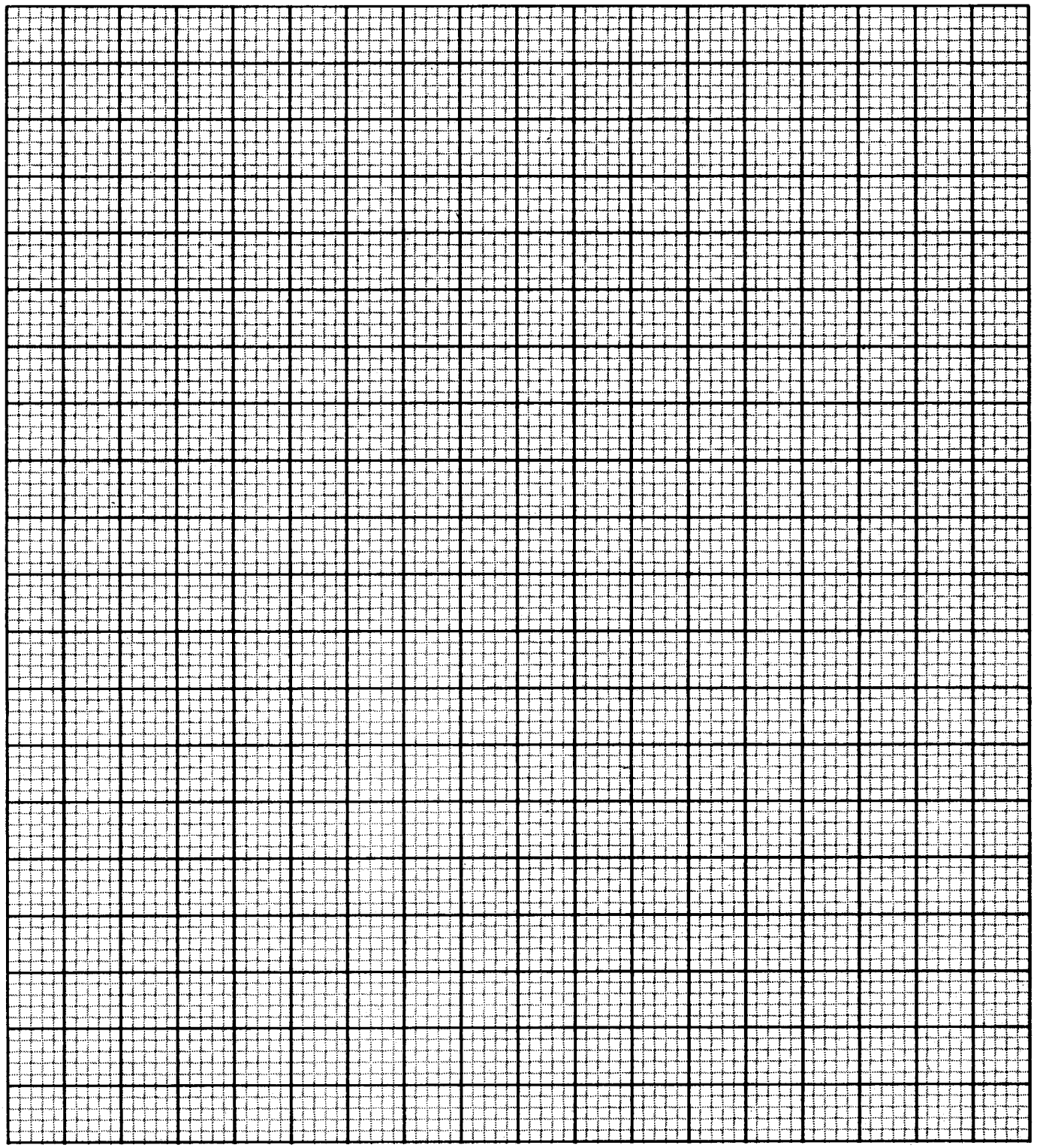
Answer sheet for Question 14 (a). **Candidate number .....**